

7.4 Device Functional Modes

The device has two functional modes: on and off. The device enters the on mode when the voltage on the VIN pin is higher than the UVLO threshold and a high logic level is applied to the EN pin. The device enters the off mode when the voltage on the VIN pin is lower than the UVLO threshold or a low logic level is applied to the EN pin.

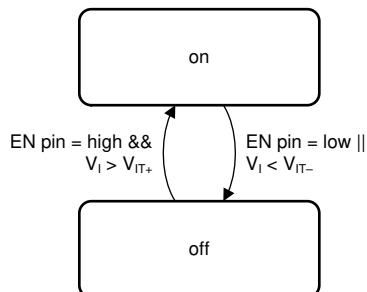


Figure 7-13. Device Functional Modes

8 Application and Implementation

Note

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes. Customers should validate and test their design implementation to confirm system functionality.

8.1 Application Information

The TPS63900 is a high efficiency, non-inverting buck-boost converter with an extremely low quiescent current, suitable for applications that need a regulated output voltage from an input supply that can be higher or lower than the output voltage. The input current limit and output voltage are set through resistors connected to the three CFGx pins.

8.2 Typical Application

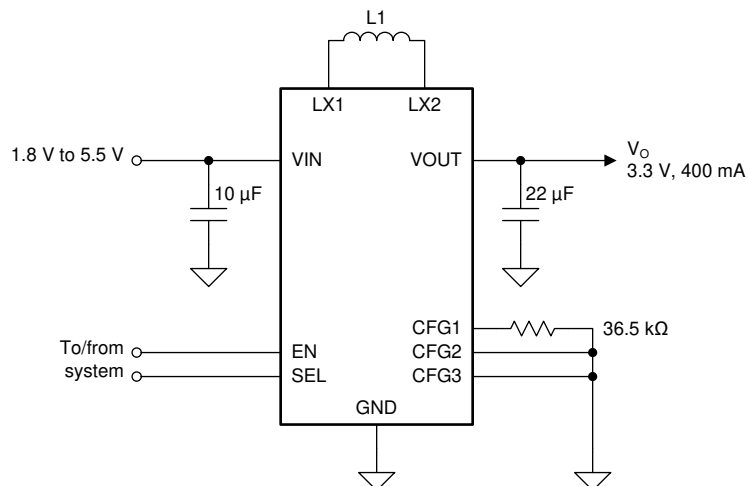


Figure 8-1. 3.3 V_{OUT} Typical Application

8.2.1 Design Requirements

The design guideline provides a component selection to operate the device within [Section 6.3](#).

Table 8-1. Matrix of Output Capacitor and Inductor Combinations

NOMINAL INDUCTOR VALUE [µH] ⁽¹⁾	NOMINAL OUTPUT CAPACITOR VALUE [µF] ⁽²⁾				
	10	22	47	100	≥ 300
2.2	+(3)	+(4)	+	+	+(5)

- (1) Inductor tolerance and current derating is anticipated. The effective inductance can vary by 20% and –30%.
- (2) Capacitance tolerance and DC bias voltage derating is anticipated. The effective capacitance can vary by 20% and –50%.
- (3) Output voltage ripple increases versus typical application.
- (4) Typical application. Other check marks indicate possible filter combinations.
- (5) Start-up time increased.

8.2.2 Detailed Design Procedure

The first step is the selection of the output filter components. To simplify this process, [Section 6.3](#) outlines minimum and maximum values for inductance and capacitance. Tolerance and derating must be taken into account when selecting nominal inductance and capacitance.

8.2.2.1 Custom Design with WEBENCH Tools

[Click here](#) to create a custom design using the TPS63900 device with the WEBENCH® Power Designer.